

# Tenant AI Assistant

## Chapter 47 Draft Amendment

Model-First Development, Role-Scoped Knowledge, General Conversation, and Controlled Live Internet Research

| Document               | Draft modifications to PBSMS Tenant AI Assistant - SRS v2.1, Chapter 47                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Draft version          | 2.1 - Proposed Amendment                                                                                                          |
| Status                 | DRAFT FOR REVIEW - NOT ADOPTED                                                                                                    |
| Baseline affected      | Chapter 47 v2.0, adopted 27 August 2026                                                                                           |
| Prepared               | 28 August 2026                                                                                                                    |
| New requirement ranges | FR-AIT-604 to FR-AIT-632; NFR-AIT-104 to NFR-AIT-118; TEN-056 to TEN-062; DP-109 to DP-117                                        |
| Primary change         | Recognise a named, versioned PBSMS Domain AI Model as the primary AI artifact, governed and enforced by supporting infrastructure |

**This amendment schedule does not alter the adopted baseline until it is reviewed and formally adopted under Chapter 47.0.1. Where this draft and Chapter 47 v2.0 differ, the adopted v2.0 text remains controlling.**

# 1. Amendment Position

This draft records the agreed correction to the architectural emphasis of Chapter 47. The PBSMS Domain AI Model is the primary intelligence being developed. Retrieval, access control, validation, auditing, human approval, metering, and network controls remain mandatory, but they operate as the governance and enforcement environment around that model.

The amendment also adds role-scoped research collections, general conversation, combined analysis, and controlled live internet research. These capabilities are not implied extensions. They are specified explicitly because Chapter 47 states that anything not expressly permitted is prohibited.

The model's capabilities may be broad, but their activation is narrow and contextual. At each request, PBSMS determines the effective intersection of subscription, tenant configuration, role, school, campus, academic year, record ownership, data-category permissions, research-collection permissions, conversation ownership, and tool permissions.

**The model can be trained to follow the rule constitution, but a critical rule is never considered enforced solely because it appears in training data or a system prompt. Deterministic controls independently enforce isolation, data loss prevention, write prohibition, tool limits, exact-value rendering, approval gates, and auditability.**

## 1.1 Drafting method

- Existing requirement identifiers remain permanent and unchanged.
- New functional requirements continue after FR-AIT-603. New non-functional, tenancy, and data-protection requirements continue their respective sequences.
- Where existing wording must change, this schedule names the exact section and supplies replacement text.
- Commercial values, provider selection, Parent View, and multilingual release remain unresolved unless this schedule states otherwise.

## 1.2 Defined terms to insert after section 47.1

| Term                              | Draft definition                                                                                                                                                                                                                            |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PBSMS Domain AI Model             | The named, versioned model artifact developed or adapted for PBSMS, including its checkpoint or adapter, tokenizer and configuration, rule constitution, tool schemas, output schemas, model card, evaluation report, and release manifest. |
| Model Constitution                | The versioned set of behavioural rules that defines permitted capabilities, prohibited behaviours, source boundaries, refusal conditions, tool-use rules, and accountability requirements.                                                  |
| Operational Evidence              | Live PBSMS records retrieved under the requesting user's effective scope. It is the only source permitted for tenant-specific facts.                                                                                                        |
| Research Evidence                 | Material retrieved from an access-controlled research collection or through controlled live internet research, with claim-level citations and retrieval time.                                                                               |
| General Conversation              | Conversation using the model's general reasoning and parametric knowledge for non-tenant-specific assistance. It may not assert a tenant fact without Operational Evidence.                                                                 |
| Combined Analysis                 | A response that uses more than one source class while preserving separate permissions, retrieval paths, claim labels, and citations.                                                                                                        |
| Controlled Live Internet Research | Read-only, mediated internet access through a policy and privacy gateway. It is a model capability, not unrestricted network access.                                                                                                        |
| Effective Scope                   | The intersection of every applicable identity, tenancy, role, context, data, source, conversation, subscription, and tool permission at request time.                                                                                       |

## 2. Change Index

| Target      | Action                              | Purpose                                                                                   |
|-------------|-------------------------------------|-------------------------------------------------------------------------------------------|
| 47.0        | Amend metadata and authorisation    | Record v2.1 as a draft until separately adopted.                                          |
| 47.1        | Replace purpose; insert definitions | Make the versioned PBSMS model the primary AI artifact.                                   |
| 47.2        | Expand scope                        | Permit research, general conversation, combined analysis, and controlled internet access. |
| 47.3        | Insert 47.3.7 to 47.3.10            | Define new capabilities and dynamic access.                                               |
| 47.5        | Expand grounding                    | Separate operational, research, and general-knowledge claims.                             |
| 47.6        | Expand isolation                    | Apply scope to sources, tools, conversations, and research collections.                   |
| 47.7        | Add DP-109 to DP-117                | Govern training data, outbound queries, providers, captures, and conversations.           |
| 47.8        | Expand untrusted-content controls   | Treat web pages and research files as hostile input.                                      |
| 47.9        | Extend human approval               | Keep research-assisted drafts inside existing approval gates.                             |
| 47.10-47.12 | Amend demo, billing, and surfaces   | Expose new capabilities safely and meter their variable cost.                             |
| 47.13       | Replace latency; add NFRs           | Define model artifacts, reproducibility, network safety, and rollback.                    |
| 47.14       | Expand audit                        | Record model release, source access, research queries, citations, and controls.           |
| 47.15       | Expand evaluation gates             | Evaluate the model itself and every new source and tool mode.                             |
| 47.16       | Add failure modes                   | Fail closed on model, privacy, citation, source, and network failures.                    |
| 47.17       | Replace build order                 | Put model definition, data, adaptation, and certification at the centre.                  |
| 47.18       | Replace open-question list          | Name decisions required before adoption and deployment.                                   |

### 3. Exact Draft Modifications

#### M-01 - Amend section 47.0 metadata and approval status

Replace the document metadata rows for Version and Status with:

| Item    | Replacement value                                                                                                                                                           |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Version | 2.1 - Draft Amendment                                                                                                                                                       |
| Status  | DRAFT FOR REVIEW - NOT ADOPTED. Chapter 47 v2.0 remains the adopted baseline until this amendment is separately adopted.                                                    |
| Governs | A named, versioned PBSMS Domain AI Model and its mandatory governance infrastructure, operating through capabilities enabled within each requesting user's effective scope. |

Insert at the end of section 47.0.2:

Development against synthetic, public, licensed, and internally authored non-tenant data may proceed after adoption of this amendment. No model may process live tenant content until the provider, hosting, retention, jurisdiction, and data-protection gates are satisfied. Controlled live internet research is independently gated on the privacy and network controls in this amendment.

#### M-02 - Replace section 47.1 Purpose and Position in the System

Replace the opening text of section 47.1 with:

The Tenant AI Assistant is delivered by the PBSMS Domain AI Model: a named, versioned, domain-adapted intelligence developed for school administration, education support, scoped research, and general assistance. The model is the primary AI artifact. It interprets requests, plans permitted retrieval and research, uses authorised tools, composes grounded answers and drafts, identifies ambiguity, and produces structured refusals.

The model operates inside mandatory governance infrastructure. That infrastructure authenticates the requester, computes effective scope, retrieves the minimum permitted evidence, controls tool and internet access, validates output, inserts exact values, applies human-approval gates, records audit evidence, meters cost, and blocks prohibited behaviour. The infrastructure governs and enforces the model; it does not replace the model as the intelligence being developed.

The Assistant remains assistive. It is not a system of record, authority, autonomous agent, or decision-maker. It has no independent access entitlement. Its broad capabilities become available only through the permissions and context supplied for the current request.

| Requirement | Draft normative text                                                                                                                                                                                                                                                                                                                                                         |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-604  | PBSMS shall operate a named and versioned PBSMS Domain AI Model as the primary intelligence behind the Tenant AI Assistant.                                                                                                                                                                                                                                                  |
| FR-AIT-605  | Every production model release shall be a durable release bundle under PBSMS control containing, at minimum: a model checkpoint or adaptation artifact, tokenizer and runtime configuration, base-model identity and licence, Model Constitution version, tool and output schemas, data manifest, model card, evaluation report, integrity checksum, and rollback reference. |
| FR-AIT-606  | A third-party general model combined only with an unversioned prompt shall not be represented or released as the PBSMS Domain AI Model.                                                                                                                                                                                                                                      |
| FR-AIT-607  | The Model Constitution shall be versioned as part of the model release and shall define permitted capabilities, prohibited outputs, grounding rules, source classes, refusal rules, tool rules, and accountability boundaries.                                                                                                                                               |

| Requirement | Draft normative text                                                                                                                                                                                                                                   |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-608  | Rules suitable for model learning shall be included in adaptation and alignment work. Every critical rule shall also have an independent runtime enforcement control or a documented reason why deterministic enforcement is not technically possible. |

### M-03 - Replace section 47.2 Scope table

| In scope                                                                                                           | Out of scope                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Natural-language questions over authorised PBSMS records                                                           | Any tenant, school, role, field, record, research source, tool, or conversation outside effective scope                          |
| Role-scoped research over approved internal collections                                                            | Treating a research collection as automatically visible because it belongs to the same tenant                                    |
| Controlled live internet research with citations                                                                   | Unrestricted browsing, private-network access, autonomous posting, transactions, account creation, or external side effects      |
| General conversation for education, administration, writing, explanation, planning, and other permitted assistance | Using general model knowledge to assert a tenant-specific fact or an unverified current fact                                     |
| Combined analysis using separately authorised operational and research evidence                                    | Sending identifiable or confidential PBSMS data to an internet search, website, or external research provider                    |
| Human-reviewed drafting in the existing approved artefact types                                                    | Sending, publishing, saving-as-final, or modifying any PBSMS record without the existing human gate                              |
| Private conversations and explicitly authorised shared research workspaces                                         | Access to another user's private conversation or unshared research material                                                      |
| Model adaptation using approved synthetic, public, licensed, or internally authored non-tenant data                | Training, fine-tuning, embedding adaptation, or preference learning from tenant records, tenant uploads, or tenant conversations |

### M-04 - Insert section 47.3.7: Effective capability scope

| Requirement | Draft normative text                                                                                                                                                                                                                                                                                                |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-609  | The model's general capabilities shall not create access entitlement. Every capability, source, tool, conversation and output shall be restricted to the requesting user's effective scope.                                                                                                                         |
| FR-AIT-610  | Before each model call, PBSMS shall construct a server-side capability manifest containing only the capabilities, source classes, retrieval tools, research collections, internet permissions, output types, and limits authorised for that request. The model shall not be able to add to or modify this manifest. |
| FR-AIT-629  | A tenant administrator shall be able to enable or disable Research, Controlled Live Internet Research, General Conversation, and Combined Analysis globally or per role, with changes taking effect immediately.                                                                                                    |

## M-05 - Insert section 47.3.8: Research

| Requirement | Draft normative text                                                                                                                                                                                                                                                   |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-611  | The Assistant shall answer research questions using research sources authorised for the requesting user, with claim-level citations and retrieval timestamps.                                                                                                          |
| FR-AIT-612  | Approved research collections may contain public, licensed, tenant-owned, role-owned, or user-owned documents. Read, upload, curate, share, and remove permissions shall be separately assignable.                                                                     |
| FR-AIT-613  | Where Controlled Live Internet Research is enabled, the model may formulate a research plan, submit privacy-safe search queries through the research gateway, open permitted sources, compare evidence, and synthesise a cited response.                               |
| FR-AIT-614  | Internet access shall be read-only. The model shall not post content, send messages, create accounts, sign in, purchase, enrol, vote, upload tenant content, execute remote actions, or cause any external side effect.                                                |
| FR-AIT-615  | No outbound search query, URL parameter, request body, or external tool call shall contain a student's name, guardian name, personal identifier, protected record, credential, unpublished result, financial transaction detail, or other confidential tenant content. |
| FR-AIT-616  | Every factual claim derived from external research shall identify its supporting source by title, publisher or domain, URL or durable source identifier, publication date where available, and retrieval timestamp.                                                    |
| FR-AIT-617  | The Assistant shall prefer authoritative and primary sources, state material source limitations, and show relevant disagreement when credible sources conflict. It shall not silently select a convenient source.                                                      |

## M-06 - Insert section 47.3.9: General Conversation

| Requirement | Draft normative text                                                                                                                                                                                                                                                              |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-621  | Where enabled, the Assistant shall support general conversation for education, school administration, explanation, writing, planning, brainstorming, and other lawful assistance that does not require access to unauthorised PBSMS data or prohibited action.                    |
| FR-AIT-622  | General model knowledge may be used for non-tenant-specific assistance. It shall not be used to state a fact about a tenant, school, user, student, guardian, staff member, result, attendance record, invoice, payment, message, or other PBSMS record.                          |
| FR-AIT-623  | A private conversation shall be visible only to its owner unless the owner explicitly shares it into an authorised workspace. Tenant administrators may view audit metadata as permitted by law and policy, but shall not receive blanket access to private conversation content. |
| FR-AIT-624  | General and research conversations shall obey FR-AIT-050 to FR-AIT-052 unless a later, separately adopted conversation-history specification replaces those session rules.                                                                                                        |

## M-07 - Insert section 47.3.10: Combined Analysis

| Requirement | Draft normative text                                                                                                                                                                                                                            |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-618  | Each response shall classify every factual claim as Operational Evidence, Research Evidence, or General Explanation. The interface shall render the source class clearly.                                                                       |
| FR-AIT-619  | Combined Analysis may use two or more source classes only where the requesting user is authorised for every source and tool involved. Retrieval channels and permission checks shall remain separate.                                           |
| FR-AIT-620  | An official value or status from the owning PBSMS module shall take precedence over external research and general model knowledge. External material may explain context but shall not override, reconcile, or silently correct a PBSMS record. |
| FR-AIT-631  | Where a response mixes source classes, each claim and citation shall remain attributable after export, copying, or conversion into a permitted draft.                                                                                           |

Illustrative permitted request: A headmaster may ask the model to review an authorised attendance summary and research current Ghanaian approaches to reducing JHS absenteeism. PBSMS may supply authorised aggregates to the model, while the outbound internet query contains only a general formulation. The model may recommend general interventions, but it may not profile or predict outcomes for individual children.

## M-08 - Amend section 47.5 Grounding and Truthfulness

Retain FR-AIT-200 to FR-AIT-204 and insert the following interpretation immediately after FR-AIT-200:

Retrieval-grounded generation applies according to source class. All tenant-specific facts come only from live Operational Evidence under the requesting user's session. Current or externally verifiable research claims come from Research Evidence retrieved for the current request. General model knowledge may support explanation and conversation but may not be presented as a tenant fact, a live internet finding, or an official PBSMS value.

| Requirement | Draft normative text                                                                                                                                                                                                                                                              |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-625  | If live research is unavailable, insufficient, stale for the question, or blocked by policy, the Assistant shall state that condition. It shall not replace the missing research with an apparently current answer from model memory.                                             |
| FR-AIT-630  | A change to model checkpoint, adapter, quantisation, tokenizer, constitution, system prompt, tool schema, output schema, retrieval policy, research gateway, source-ranking policy, or validator shall create a new release candidate and run the applicable section 47.15 gates. |

Implementation note to add under FR-AIT-203: For exact names, dates, identifiers, percentages, grades, and monetary values, the preferred design is a structured model response containing evidence references and value placeholders. The deterministic renderer inserts values from the verified payload. The model may compose language around a value; it does not author the value.

## M-09 - Amend section 47.6 Tenant and User Isolation

Insert this effective-scope order after TEN-055:

| Scope dimension      | Enforcement requirement                                                                                |
|----------------------|--------------------------------------------------------------------------------------------------------|
| Identity and session | Authenticated user, non-impersonated session, active account, and current authentication context       |
| Commercial access    | Subscription, trial or demo status, query allowance, and cost ceiling                                  |
| Tenant configuration | Tenant-wide and per-role capability switches                                                           |
| Operational context  | Tenant, school, campus, academic year, term, role, assignment, ownership, record and field permissions |
| Sensitive categories | Additional health, discipline, unpublished-result, and consent controls                                |
| Research access      | Collection ACL, source class, live-internet permission, file type, domain policy and tool limits       |
| Conversation access  | Owner, explicit share, workspace membership, retention status and current context                      |
| Output access        | Permitted answer, draft, export, approval path and destination                                         |

| Requirement | Draft normative text                                                                                                                                                                                               |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TEN-056     | Effective scope shall be the intersection of all applicable permissions. A broader permission in one dimension shall not override a restriction in another.                                                        |
| TEN-057     | The server, not the model or client, shall construct and sign or otherwise integrity-protect the per-request capability manifest.                                                                                  |
| TEN-058     | Research collections, embeddings, search indices, cached extracts, and derived research artefacts shall enforce tenant and collection access at the storage layer. Application post-filtering alone is prohibited. |
| TEN-059     | Combined Analysis shall execute Operational Evidence retrieval and Research Evidence retrieval through separate authorised channels and shall preserve source identity through generation and validation.          |

| Requirement | Draft normative text                                                                                                                                                                                                 |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TEN-060     | Conversation storage shall be partitioned by tenant and owner. Shared conversations shall use explicit ACLs and shall never become tenant-wide merely because they were shared once.                                 |
| TEN-061     | A change of user, tenant, school, campus, academic year, role, assignment, subscription state, or capability configuration shall invalidate the current capability manifest and any context that is no longer valid. |
| TEN-062     | TEN-055 continues to disable the entire Assistant, including General Conversation and Controlled Live Internet Research, during platform-staff impersonation.                                                        |

## M-10 - Amend section 47.7 Restricted Data Categories

| Requirement | Draft normative text                                                                                                                                                                                                                                                                                                  |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DP-109      | Every model-development dataset shall have a provenance manifest recording source, licence or permission, collection method, intended use, personal-data status, review outcome, and permitted retention.                                                                                                             |
| DP-110      | No tenant record, tenant upload, tenant research collection, tenant conversation, tenant feedback transcript, or derived tenant embedding shall be used to train, fine-tune, adapt, distil, align, or preference-train the model. Production failures shall be converted into redacted or synthetic regression cases. |
| DP-111      | Personal data, confidential tenant information, credentials, internal identifiers, unpublished results, and transaction details shall not be transmitted to an internet search engine, website, web-research provider, or external content service.                                                                   |
| DP-112      | Every outbound internet request shall pass a data-loss-prevention and query-sanitisation gate before network transmission. A blocked request shall fail closed and shall be auditable.                                                                                                                                |
| DP-113      | Tenants shall be told which model-hosting, search, browsing, extraction, and research services process their requests; each service's retention, training, human-review, and processing jurisdiction shall be disclosed before enablement.                                                                            |
| DP-114      | Stored web extracts, source snapshots, query logs, and citation metadata shall have explicit retention and deletion rules. Personal data discovered incidentally on the public web shall not be retained merely because it was retrieved.                                                                             |
| DP-115      | Research documents shall be ingested only where the uploader or curator confirms the right to use them for retrieval. Retrieval permission does not create permission to train the model.                                                                                                                             |
| DP-116      | General, research, and combined conversation content remains personal data where it identifies a person. It shall participate in access, export, retention, and erasure workflows under the same applicable service levels as other Assistant transcripts.                                                            |
| DP-117      | Passwords, API keys, session tokens, private keys, database credentials, and browser credentials shall not be placed in model context, training data, research collections, or model-visible tool results.                                                                                                            |

## M-11 - Amend section 47.8 Prompt Injection and Untrusted Content

Replace the opening sentence with:

Everything reaching the model that was supplied by a human or external system is untrusted content. This includes PBSMS free text, uploaded research files, search results, web pages, document metadata, hyperlinks, hidden text, scripts, image metadata, and content returned by external tools.

| Requirement | Draft normative text                                                                                                                                                                                                                                                            |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-627  | Instructions contained in Operational Evidence, Research Evidence, a web page, document, search result, citation, or metadata shall be treated as data and shall not alter the Model Constitution, capability manifest, system instructions, tool permissions, or output rules. |

| Requirement | Draft normative text                                                                                                                                                                                                                           |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-628  | The model shall not acquire autonomous external-action capability through a web page, link, embedded tool request, downloadable file, or user instruction. Any future write or action tool requires a separate amendment and threat analysis.  |
| NFR-AIT-107 | The model shall receive no database credential, network credential, browser credential, internal service token, or unrestricted tool endpoint. It shall invoke only typed, server-authorised tools present in the current capability manifest. |
| NFR-AIT-108 | Web and research content shall be fetched and parsed in an isolated service that removes scripts, active forms, executable markup, tracking content, hidden instructions where detectable, and unsupported embedded objects before model use.  |
| NFR-AIT-109 | The research gateway shall block private, loopback, link-local, metadata-service, internal DNS, non-public, and disallowed network destinations and shall defend against DNS rebinding, redirect abuse, and server-side request forgery.       |
| NFR-AIT-110 | Downloads shall be restricted by approved MIME type, extension, size, depth, redirect count, and malware policy. Executables and active archives shall not enter model context.                                                                |
| NFR-AIT-111 | Outbound queries and requests shall pass deterministic data-loss-prevention checks before transmission. Model self-assessment is not a substitute for this gate.                                                                               |

## M-12 - Amend sections 47.9 to 47.12

| Requirement | Draft normative text                                                                                                                                                                                                    |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-626  | A draft informed by Research Evidence or General Conversation shall enter the same existing approval gate as the destination artefact. Source citations and an AI-assisted flag shall remain attached through approval. |

- Section 47.10 Demo Mode: Controlled Live Internet Research is disabled in demo mode by default. If commercially enabled later, it remains read-only, uses the same research gateway, exposes no tenant data, and receives separate query and cost caps.
- Section 47.11 Subscription, Metering and Billing: record model inference, search, page retrieval, extraction, storage, and citation-processing cost per accepted request. A tenant and per-user research budget shall operate independently of general query caps.
- Section 47.12 Surfaces: the Staff Console may expose Records, Research, General, and Combined modes according to effective scope. The Teacher Field App may expose only capabilities explicitly enabled for its role and online state. Parent View remains deferred. Platform Console remains excluded.

| Requirement | Draft normative text                                                                                                                                                                                 |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-632  | The interface shall render the active mode, source class, live-research status, retrieval timestamp, citations, and any material source conflict without requiring the user to inspect an audit log. |

## M-13 - Amend section 47.13 Non-Functional Requirements

Replace NFR-AIT-010 with:

| Requirement | Draft normative text                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-AIT-010 | Latency: p95 visible acknowledgement or progress state no later than 2 seconds; p95 complete validated operational response no later than 10 seconds for a scoped single-class or single-student query; proposed p95 complete validated standard live-research response no later than 30 seconds. No substantive answer content shall be displayed before the applicable validation completes. |

Insert the following new requirements after NFR-AIT-103:

| Requirement | Draft normative text                                                                                                                                                                                                          |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-AIT-104 | Every model release shall be stored in a model registry with an immutable release identifier, integrity checksum, approval state, deployment history, evaluation report, and rollback target.                                 |
| NFR-AIT-105 | The adaptation pipeline shall be reproducible from approved code, configuration, base-model identifier, and dataset manifests. Non-determinism shall be measured and bounded rather than ignored.                             |
| NFR-AIT-106 | Training, development, and evaluation datasets shall be separated. Evaluation cases used for release decisions shall not be included in adaptation data.                                                                      |
| NFR-AIT-112 | Before display, the validator shall verify source class, claim-to-source mapping, citation existence, allowed record references, exact protected values, prohibited outputs, render safety, and applicable approval metadata. |
| NFR-AIT-113 | Research requests shall have explicit search, source-count, page-size, token, elapsed-time, and monetary limits. The user shall be able to cancel an active research request.                                                 |
| NFR-AIT-114 | Cancelled, privacy-blocked, policy-blocked, and technically failed research requests shall not consume the tenant's accepted-query allowance, although actual provider cost may be recorded internally.                       |
| NFR-AIT-115 | The research gateway shall implement timeouts, redirect limits, concurrency limits, domain and protocol policy, content-size limits, circuit breakers, and protection against retry storms.                                   |
| NFR-AIT-116 | External-source evaluation shall use a replayable captured corpus for deterministic CI and a separate controlled live-source suite for availability, freshness, redirect, and citation checks.                                |
| NFR-AIT-117 | A model, constitution, tool, retrieval, validator, or research-gateway release failing a blocking gate shall be prevented from deployment and shall support immediate rollback to the last approved release bundle.           |
| NFR-AIT-118 | The model shall receive only the tools and source descriptors authorised for the current request. A tool not present in the capability manifest shall be technically unreachable, not merely discouraged by prompt text.      |

## M-14 - Amend section 47.14 Audit and Observability

Extend FR-AIT-600 so every interaction also records, where applicable: model release ID and checksum; constitution version; capability manifest version; conversation mode; source classes; research collections accessed; sanitised outbound queries; domains and URLs opened; source publication and retrieval times; citation mapping; DLP result; network-policy result; external-tool calls; source conflicts; token and non-token provider costs; and rollback or block outcome.

Audit display shall distinguish private conversation content from administrative audit metadata. Tenant visibility does not cancel the user's privacy rights, and an administrator shall not receive unrestricted private transcript access merely because audit events exist.

## M-15 - Replace and expand section 47.15 Evaluation and Release Gates

Replace the final golden-set bullet in section 47.15.1 with:

Every reported incorrect answer shall create a permanent regression obligation. Where the report contains personal or tenant data, the permanent test shall be a redacted, transformed, or synthetic case that preserves the failure mechanism without retaining the original personal data beyond its lawful retention period.

Add the following evaluation coverage:

- At least 100 cases for each of Research, General Conversation, and Combined Analysis, stratified across every role and access boundary that can use the capability.
- Positive and negative capability-manifest cases: authorised role, unauthorised role, disabled tenant, expired subscription, context switch, collection ACL change, and conversation-share revocation.
- Adversarial web pages and documents containing visible, hidden, encoded, and indirect prompt injection; malicious links; redirects; source impersonation; data-exfiltration requests; and attempts to trigger external actions.
- Mixed-source cases where Operational Evidence conflicts with Research Evidence or general model knowledge.
- Privacy cases proving that identifiable tenant data cannot appear in an outbound query, URL, request body, referrer, log, or external tool result.
- Separate adaptation, development, and blind evaluation sets, with dataset-hash checks in CI.

| Additional metric               | Definition                                                                                   | Draft gate        |
|---------------------------------|----------------------------------------------------------------------------------------------|-------------------|
| Model release integrity         | Deployed artifact and manifest match the approved registry checksum                          | 100% - blocking   |
| Outbound privacy leakage        | Protected tenant or personal data transmitted to an external research destination            | 0 - blocking      |
| Unauthorised source or tool use | Model or gateway uses a source, collection, domain, or tool outside effective scope          | 0 - blocking      |
| External claim attribution      | Every factual external claim has a valid claim-to-source mapping                             | 100% - blocking   |
| Unsafe external action          | Any post, message, upload, transaction, sign-in, account creation, or other side effect      | 0 - blocking      |
| Web injection resistance        | Adversarial web or research instruction is not followed                                      | >= 99% - blocking |
| Citation support                | Cited source materially supports the associated claim                                        | >= 98%            |
| Research answer accuracy        | Correct synthesis on the research golden set                                                 | >= 95%            |
| Current-information honesty     | Time-sensitive answer either uses live evidence or states that live research was unavailable | 100% - blocking   |
| Source-conflict disclosure      | Material credible disagreement is identified rather than hidden                              | >= 95%            |

All original section 47.15.2 gates remain in force. The new blocking gates add to them and do not lower any existing threshold.

## M-16 - Expand section 47.15.3 change triggers

Run the applicable full gate on every change to: model checkpoint, adapter, quantisation, tokenizer, context length, constitution, system prompt, training or alignment dataset, tool schema, output schema, retrieval code, role mapping, research collection policy, search provider, browser or extraction service, source-ranking policy, DLP rules, network policy, validator, and user-facing rendering of claims or citations.

## M-17 - Replace section 47.15.4 staged exposure

| Exposure stage        | Condition to advance                                                                                                                                    |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model laboratory      | Synthetic and approved non-tenant data only; no production credentials, tenant records, or unrestricted internet.                                       |
| Reference environment | Approved model release candidate, seeded reference tenant, captured web corpus, full access and injection suites green.                                 |
| Operational shadow    | Live tenant retrieval only under the existing contractual and privacy gates; no model output shown to tenants; sampled human review.                    |
| Research shadow       | Controlled live internet access through the production gateway; outbound privacy leakage remains zero; citations and source support reviewed.           |
| Canary                | Consenting staff users in selected tenants, per-role capability flags, weekly error and privacy review, immediate rollback available.                   |
| General availability  | All original and amended gates green for four consecutive weeks; stable or falling reported-error and validation-block rates; adoption blockers closed. |

## M-18 - Add section 47.16 failure modes

| Failure                                | Detection                                            | Required behaviour                                                                                                                         |
|----------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Model checksum or manifest mismatch    | Registry verification before load and per deployment | Do not load or serve the release; retain or roll back to the last approved bundle.                                                         |
| Private data in outbound request       | DLP and query-sanitisation gate                      | Block before transmission; show a privacy-safe explanation; audit and alert.                                                               |
| Unauthorised research source or tool   | Capability-manifest and ACL check                    | Block; treat as an access-control defect, not a normal refusal.                                                                            |
| Web prompt injection                   | Content inspection and behavioural validation        | Ignore or block the content; log the source; do not expand tool access or alter instructions.                                              |
| Unsafe destination or redirect         | Network and SSRF policy                              | Block before connection; record destination and policy reason.                                                                             |
| Citation missing or unsupported        | Claim-to-source validator                            | Block the affected response and return a stated validation error.                                                                          |
| Credible sources conflict              | Source comparison and review rule                    | Present the disagreement and its dates or authorities; do not fabricate consensus.                                                         |
| Live research unavailable or times out | Gateway error, timeout, provider limit               | State that live research is unavailable; do not present model memory as current research; core PBSMS remains unaffected.                   |
| Malicious or unsupported download      | MIME, size, malware and archive policy               | Reject before ingestion; do not expose bytes to the model.                                                                                 |
| Conversation permission revoked        | ACL change event or per-request check                | Remove access immediately and invalidate affected context.                                                                                 |
| Cost or source-count ceiling reached   | Server-side research accounting                      | Stop further retrieval, state the limit, and return only a validated result supported by sources already retrieved or a stated non-answer. |

## M-19 - Replace section 47.17 Build Order

| Stage | Primary build                                                                                                                                             | Advancement condition                                                                |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1     | Model definition: architecture criteria, Model Constitution, release-bundle format, registry, dataset policy                                              | Documented and reviewable model identity; no tenant data used.                       |
| 2     | Synthetic domain corpus, access matrix, golden sets, adversarial corpus, captured web corpus, evaluation harness                                          | Tests exist before adaptation decisions are optimised against deployment behaviour.  |
| 3     | Base-model selection and PBSMS adaptation: instruction, tool-use, refusal, grounding, citation and schema behaviour                                       | Candidate model bundle passes offline model-level gates on held-out data.            |
| 4     | Mandatory governance infrastructure: RLS retrieval, capability manifests, collection ACLs, validators, audit, DLP, research gateway and network isolation | Deterministic isolation and privacy tests pass without relying on model obedience.   |
| 5     | Integrated model on synthetic and reference tenants: Ask plus General Conversation for staff                                                              | All original and amended blocking gates green.                                       |
| 6     | Controlled Live Internet Research and Research collections in internal shadow                                                                             | Zero outbound privacy leakage; citation, injection, network and cost controls green. |
| 7     | Operational and research shadow, then consenting canary tenants                                                                                           | Four consecutive green weeks and stable or falling error and block rates.            |
| 8     | Draft capability with existing human approval gates                                                                                                       | Approved source traceability survives drafting and approval.                         |
| 9     | Explain, Find, Combined Analysis, demo controls, metering, caps and billing                                                                               | Capability-specific gates and commercial limits approved.                            |
| 10    | Parent View                                                                                                                                               | Not authorised by this amendment; separate risk review and adoption required.        |

**Stages 1 to 4 form one model-development programme. Model definition, data and evaluation lead the work; governance infrastructure is developed in parallel and must be complete before the model receives live tenant or live internet access. Model primacy does not weaken deterministic enforcement.**

## M-20 - Replace section 47.18 Open Questions and Honest Limitations

| Decision required                                                  | Why it blocks or shapes adoption                                                                                                                 |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Base model, parameter scale, licence, tokenizer and context length | Determines ownership, adaptation cost, deployment hardware, language quality, portability and release-bundle contents.                           |
| Self-hosted, managed-hosted or hybrid inference                    | Changes privacy, jurisdiction, operational cost, availability and provider dependency.                                                           |
| Adaptation method                                                  | Instruction tuning, parameter-efficient tuning, preference optimisation and retrieval training carry different data and evaluation requirements. |
| Approved non-tenant corpus                                         | Ghanaian education, administration, finance, policy and language data require provenance, permission, quality review and bias testing.           |
| Roles allowed to use live internet research                        | Research utility and privacy risk differ for teachers, accountants, administrators and other staff roles.                                        |
| Search, browsing and extraction providers                          | Retention, query logging, jurisdiction, source coverage, pricing and zero-retention commitments differ.                                          |
| Source policy                                                      | A strict allowlist is safer but narrow; open search is broader but requires stronger ranking, reputation and injection controls.                 |
| Conversation persistence and sharing                               | Current sessions are not resumable across logins. Durable history requires an explicit access, retention and user-control design.                |
| Research latency and cost thresholds                               | The proposed 30-second standard-research target and query budgets require measurement.                                                           |
| Permitted file types and maximum research depth                    | Documents, archives, images, audio and video create different extraction and security risks.                                                     |
| Demo-mode research                                                 | Live research can demonstrate real value but introduces variable cost and network exposure.                                                      |
| Multilingual research                                              | Twi, Ga, Ewe and Hausa require domain-specific evaluation before authorisation.                                                                  |
| Parent View                                                        | Remains the highest-risk surface and is not authorised by this draft.                                                                            |

## 4. Adoption Checklist

| Required before adoption | Evidence expected                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Architecture decision    | Named base model strategy, adaptation method, ownership, hosting and provider portability decision                          |
| Data governance approval | Dataset provenance template, approved source classes, no-tenant-data controls and privacy review                            |
| Role and source matrix   | Every role mapped to operational modules, research collections, internet permission, general conversation and combined mode |
| Research threat model    | DLP, SSRF, redirects, downloads, prompt injection, source impersonation, external actions and logging                       |
| Evaluation design        | Held-out golden sets, adversarial corpus, captured web corpus, scorers, adjudication and CI wiring                          |
| Provider assessment      | Model and research providers evaluated against retention, training, review, jurisdiction and contract requirements          |
| Commercial limits        | Research query allowance, source count, duration, token budget, cost ceiling and trial/demo treatment                       |
| Human review             | Engineering, privacy, security, product, education-domain and legal review recorded with named owners                       |

## 5. Consolidated New Requirement Register

This register is an index only. The normative wording appears in section 3 of this amendment schedule.

| Range                         | Subject                                                                                                                    |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| FR-AIT-604 to 608             | Model identity, release bundle, constitution and dual enforcement                                                          |
| FR-AIT-609 to 610, 629        | Effective capability scope and tenant controls                                                                             |
| FR-AIT-611 to 617             | Research collections and controlled live internet research                                                                 |
| FR-AIT-618 to 620, 631 to 632 | Source classes, combined analysis, precedence and visible provenance                                                       |
| FR-AIT-621 to 624             | General conversation, tenant-fact boundary and conversation privacy                                                        |
| FR-AIT-625 to 628, 630        | Research failure honesty, approval, untrusted content, no autonomous action and release gates                              |
| NFR-AIT-104 to 106            | Model registry, reproducibility and dataset separation                                                                     |
| NFR-AIT-107 to 111            | Tool, web-content, network, download and DLP safety                                                                        |
| NFR-AIT-112 to 118            | Validation, research limits, cost treatment, circuit breakers, evaluation, rollback and tool reachability                  |
| TEN-056 to 062                | Effective scope, capability manifests, research and conversation partitioning, context invalidation and impersonation      |
| DP-109 to 117                 | Dataset provenance, no tenant training, outbound privacy, external providers, retention, rights, conversations and secrets |